

Bayes Theorem

15 June 2023 18:16

Host

Bayes' Theorem is a fundamental concept in the field of probability and statistics that describes how to update the probabilities of hypotheses when given evidence. It's used in a wide variety of fields, including machine learning, statistics, and game theory.

The theorem is named after Thomas Bayes, who introduced an early version of the rule in his work on probability theory. Bayes' Theorem provides a way to revise existing predictions or theories (update probabilities) given new evidence.



$$\rightarrow \boxed{P(A|B)} = \boxed{P(A)} \times \frac{\boxed{P(B|A)}}{\boxed{P(B)}}$$

posterior prior likelihood / marginal evidence

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

gender

survived

M → 0

M → 0

F → 1

F → 0

M → 1

$P(A|B) = \frac{P(B)}{P(A)}$

$\{M\} \rightarrow \{0, 1\} \rightarrow \frac{2}{3}$

$P(\underline{0} | \underline{M}) = \frac{P(\underline{M} | \underline{0}) P(\underline{0})}{P(\underline{M})} \rightarrow \frac{\frac{2}{5}}{\frac{3}{5}} = \frac{2}{3}$

$P(1 | M) = \frac{P(M | 1) P(1)}{P(M)} \rightarrow \frac{\frac{1}{2} \times \frac{2}{5}}{\frac{3}{5}} = \frac{1}{3}$

$P(0 | M) = \frac{2}{3}$

$P(1 | M) = \frac{1}{3}$

dead > alive

$\frac{1}{2} \times \frac{2}{5} = \frac{1}{5}$

$\frac{1}{5} \times \frac{2}{3} = \frac{2}{15}$

$\frac{2}{15} \times \frac{3}{5} = \frac{2}{25}$

$\frac{2}{25} \times \frac{3}{5} = \frac{2}{125}$

$\frac{2}{125} \times \frac{3}{5} = \frac{2}{625}$

$\frac{2}{625} \times \frac{3}{5} = \frac{2}{3125}$

$\frac{2}{3125} \times \frac{3}{5} = \frac{2}{15625}$

$\frac{2}{15625} \times \frac{3}{5} = \frac{2}{78125}$

$\frac{2}{78125} \times \frac{3}{5} = \frac{2}{390625}$

$\frac{2}{390625} \times \frac{3}{5} = \frac{2}{1953125}$

$\frac{2}{1953125} \times \frac{3}{5} = \frac{2}{9765625}$

$\frac{2}{9765625} \times \frac{3}{5} = \frac{2}{48828125}$

$\frac{2}{48828125} \times \frac{3}{5} = \frac{2}{244140625}$

$\frac{2}{244140625} \times \frac{3}{5} = \frac{2}{1220703125}$

$\frac{2}{1220703125} \times \frac{3}{5} = \frac{2}{6103515625}$

$\frac{2}{6103515625} \times \frac{3}{5} = \frac{2}{30517578125}$

$\frac{2}{30517578125} \times \frac{3}{5} = \frac{2}{152587890625}$

$\frac{2}{152587890625} \times \frac{3}{5} = \frac{2}{762939453125}$

$\frac{2}{762939453125} \times \frac{3}{5} = \frac{2}{3814697265625}$

$\frac{2}{3814697265625} \times \frac{3}{5} = \frac{2}{19073486328125}$

$\frac{2}{19073486328125} \times \frac{3}{5} = \frac{2}{95367431640625}$

$\frac{2}{95367431640625} \times \frac{3}{5} = \frac{2}{476837158203125}$

$\frac{2}{476837158203125} \times \frac{3}{5} = \frac{2}{2384185791015625}$

$\frac{2}{2384185791015625} \times \frac{3}{5} = \frac{2}{11920928955078125}$

$\frac{2}{11920928955078125} \times \frac{3}{5} = \frac{2}{59604644775390625}$

$\frac{2}{59604644775390625} \times \frac{3}{5} = \frac{2}{298023223876953125}$

$\frac{2}{298023223876953125} \times \frac{3}{5} = \frac{2}{1490116119384765625}$

$\frac{2}{1490116119384765625} \times \frac{3}{5} = \frac{2}{7450580596923828125}$

$\frac{2}{7450580596923828125} \times \frac{3}{5} = \frac{2}{37252902984619140625}$

$\frac{2}{37252902984619140625} \times \frac{3}{5} = \frac{2}{186264514923095703125}$

$\frac{2}{186264514923095703125} \times \frac{3}{5} = \frac{2}{931322574615478515625}$

$\frac{2}{931322574615478515625} \times \frac{3}{5} = \frac{2}{4656612873077392578125}$

$\frac{2}{4656612873077392578125} \times \frac{3}{5} = \frac{2}{23283064365386962890625}$

$\frac{2}{23283064365386962890625} \times \frac{3}{5} = \frac{2}{116415321826934814453125}$

$\frac{2}{116415321826934814453125} \times \frac{3}{5} = \frac{2}{582076609134674072265625}$

$\frac{2}{582076609134674072265625} \times \frac{3}{5} = \frac{2}{2910383045673370361328125}$

$\frac{2}{2910383045673370361328125} \times \frac{3}{5} = \frac{2}{14551915228366851806640625}$

$\frac{2}{14551915228366851806640625} \times \frac{3}{5} = \frac{2}{72759576141834259033203125}$

$\frac{2}{72759576141834259033203125} \times \frac{3}{5} = \frac{2}{363797880709171295166015625}$

$\frac{2}{363797880709171295166015625} \times \frac{3}{5} = \frac{2}{1818989403545856475830078125}$

$\frac{2}{1818989403545856475830078125} \times \frac{3}{5} = \frac{2}{9094947017729282379150390625}$

$\frac{2}{9094947017729282379150390625} \times \frac{3}{5} = \frac{2}{45474735088646411895751953125}$

$\frac{2}{45474735088646411895751953125} \times \frac{3}{5} = \frac{2}{227373675443232059478759765625}$

$\frac{2}{227373675443232059478759765625} \times \frac{3}{5} = \frac{2}{1136868377216160297393798828125}$

$\frac{2}{1136868377216160297393798828125} \times \frac{3}{5} = \frac{2}{5684341886080801486968994140625}$

$\frac{2}{5684341886080801486968994140625} \times \frac{3}{5} = \frac{2}{28421709430404007434844970703125}$

$\frac{2}{28421709430404007434844970703125} \times \frac{3}{5} = \frac{2}{142108547152020037174224853515625}$

$\frac{2}{142108547152020037174224853515625} \times \frac{3}{5} = \frac{2}{710542735760100185871124267578125}$

$\frac{2}{710542735760100185871124267578125} \times \frac{3}{5} = \frac{2}{3552713678800500929355621337890625}$

$\frac{2}{3552713678800500929355621337890625} \times \frac{3}{5} = \frac{2}{17763568394002504646778106689453125}$

$\frac{2}{17763568394002504646778106689453125} \times \frac{3}{5} = \frac{2}{88817841970012523233890533447265625}$

$\frac{2}{88817841970012523233890533447265625} \times \frac{3}{5} = \frac{2}{444089209850062616169452667236328125}$

$\frac{2}{444089209850062616169452667236328125} \times \frac{3}{5} = \frac{2}{2220446049250313080847263336181640625}$

$\frac{2}{2220446049250313080847263336181640625} \times \frac{3}{5} = \frac{2}{11102230246251565404236316680908203125}$

$\frac{2}{11102230246251565404236316680908203125} \times \frac{3}{5} = \frac{2}{55511151231257827021181583404541015625}$

$\frac{2}{55511151231257827021181583404541015625} \times \frac{3}{5} = \frac{2}{277555756156289135105907917022705078125}$

$\frac{2}{277555756156289135105907917022705078125} \times \frac{3}{5} = \frac{2}{1387778780781445675529539585113525390625}$

$\frac{2}{1387778780781445675529539585113525390625} \times \frac{3}{5} = \frac{2}{6938893903907228377647697925567626953125}$

$\frac{2}{6938893903907228377647697925567626953125} \times \frac{3}{5} = \frac{2}{34694469519536141888238489627838134765625}$

$\frac{2}{34694469519536141888238489627838134765625} \times \frac{3}{5} = \frac{2}{173472347597680709441192448139190673828125}$

$\frac{2}{173472347597680709441192448139190673828125} \times \frac{3}{5} = \frac{2}{867361737988403547205962240695953369140625}$

$\frac{2}{867361737988403547205962240695953369140625} \times \frac{3}{5} = \frac{2}{4336808689942017736029811203479766845703125}$

$\frac{2}{4336808689942017736029811203479766845703125} \times \frac{3}{5} = \frac{2}{21684043449710088680149056017398834228515625}$

$\frac{2}{21684043449710088680149056017398834228515625} \times \frac{3}{5} = \frac{2}{108420217248550443400745280086994171142578125}$

$\frac{2}{108420217248550443400745280086994171142578125} \times \frac{3}{5} = \frac{2}{542101086242752217003726400434970855712890625}$

$\frac{2}{542101086242752217003726400434970855712890625} \times \frac{3}{5} = \frac{2}{2710505431213761085018632002174854278564453125}$

$\frac{2}{2710505431213761085018632002174854278564453125} \times \frac{3}{5} = \frac{2}{13552527156068805425093160010874271392822265625}$

$\frac{2}{13552527156068805425093160010874271392822265625} \times \frac{3}{5} = \frac{2}{67762635780344027125465800054371356964111328125}$

$\frac{2}{67762635780344027125465800054371356964111328125} \times \frac{3}{5} = \frac{2}{338813178901720135627329000271856784820556640625}$

$\frac{2}{338813178901720135627329000271856784820556640625} \times \frac{3}{5} = \frac{2}{1694065894508600678136645001359283924102783203125}$

$\frac{2}{1694065894508600678136645001359283924102783203125} \times \frac{3}{5} = \frac{2}{8470329472543003390683225006796419620513916015625}$

$\frac{2}{8470329472543003390683225006796419620513916015625} \times \frac{3}{5} = \frac{2}{42351647362715016953416125033982098102569580078125}$

$\frac{2}{42351647362715016953416125033982098102569580078125} \times \frac{3}{5} = \frac{2}{211758236813575084767080625169910490512847900390625}$

$\frac{2}{211758236813575084767080625169910490512847900390625} \times \frac{3}{5} = \frac{2}{1058791184067875423835403125849552452564239501953125}$

$\frac{2}{1058791184067875423835403125849552452564239501953125} \times \frac{3}{5} = \frac{2}{5293955920339377119177015629247762262821197509765625}$

$\frac{2}{5293955920339377119177015629247762262821197509765625} \times \frac{3}{5} = \frac{2}{26469779601696885595885078146238811314105987548828125}$

$\frac{2}{26469779601696885595885078146238811314105987548828125} \times \frac{3}{5} = \frac{2}{132348898008484427979425390731194056570529937744140625}$

$\frac{2}{132348898008484427979425390731194056570529937744140625} \times \frac{3}{5} = \frac{2}{661744490042422139897126953655970282852649688720703125}$

$\frac{2}{661744490042422139897126953655970282852649688720703125} \times \frac{3}{5} = \frac{2}{3308722450212110699485634768279851414263248443603515625}$

$\frac{2}{3308722450212110699485634768279851414263248443603515625} \times \frac{3}{5} = \frac{2}{16543612251060553497428173841399257071316242218017578125}$

$\frac{2}{16543612251060553497428173841399257071316242218017578125} \times \frac{3}{5} = \frac{2}{82718061255302767487140869206996285356581211090087890625}$

$\frac{2}{82718061255302767487140869206996285356581211090087890625} \times \frac{3}{5} = \frac{2}{413590306276513837435704346034981426782906055450439453125}$

$\frac{2}{413590306276513837435704346034981426782906055450439453125} \times \frac{3}{5} = \frac{2}{2067951531382569187178521730174907133914530277252197265625}$

$\frac{2}{2067951531382569187178521730174907133914530277252197265625} \times \frac{3}{5} = \frac{2}{10339757656912845935892608650874535669572651386260986328125}$

$\frac{2}{10339757656912845935892608650874535669572651386260986328125} \times \frac{3}{5} = \frac{2}{51698788284564229679463043254372678347863256931304931640625}$

$\frac{2}{51698788284564229679463043254372678347863256931304931640625} \times \frac{3}{5} = \frac{2}{258493941422821148397315216271863391739316284656524658203125}$

$\frac{2}{258493941422821148397315216271863391739316284656524658203125} \times \frac{3}{5} = \frac{2}{1292469707114105741986576081359316958696581423282623291015625}$

$\frac{2}{1292469707114105741986576081359316958696581423282623291015625} \times \frac{3}{5} = \frac{2}{6462348535570528709932880406796584793482907116413116455078125}$

$\frac{2}{6462348535570528709932880406796584793482907116413116455078125} \times \frac{3}{5} = \frac{2}{32311742677852643549664402033982923967414535582065582275390625}$

$\frac{2}{32311742677852643549664402033982923967414535582065582275390625} \times \frac{3}{5} = \frac{2}{161558713389263217748322010169914619837072677910327911376953125}$

$\frac{2}{161558713389263217748322010169914619837072677910327911376953125} \times \frac{3}{5} = \frac{2}{807793566946316088741610050849573099185363389551639556884765625}$

$\frac{2}{807793566946316088741610050849573099185363389551639556884765625} \times \frac{3}{5} = \frac{2}{4038967834731580443708050254247865495926816947758197784423828125}$

$\frac{2}{4038967834731580443708050254247865495926816947758197784423828125} \times \frac{3}{5} = \frac{2}{20194839173657902218540251271239327479634084738790988922119140625}$

$\frac{2}{20194839173657902218540251271239327479634084738790988922119140625} \times \frac{3}{5} = \frac{2}{100974195868289511092701256356196637398170423693954944610595703125}$

$\frac{2}{100974195868289511092701256356196637398170423693954944610595703125} \times \frac{3}{5} = \frac{2}{504870979341447555463506281780983186990852118469774723052978515625}$

$\frac{2}{504870979341447555463506281780983186990852118469774723052978515625} \times \frac{3}{5} = \frac{2}{2524354896707237777317531408904915934954260592348873615264892578125}$

$\frac{2}{2524354896707237777317531408904915934954260592348873615264892578125} \times \frac{3}{5} = \frac{2}{12621774483536188886587657044524579674771302961744368076324462890625}$

$\frac{2}{12621774483536188886587657044524579674771302961744368076324462890625} \times \frac{3}{5} = \frac{2}{63108872417680944432938285222622898373856514808721840381622314453125}$

$\frac{2}{63108872417680944432938285222622898373856514808721840381622314453125} \times \frac{3}{5} = \frac{2}{315544362088404722164691426113114491869282574043609201908111572265625}$

$\frac{2}{315544362088404722164691426113114491869282574043609201908111572265625} \times \frac{3}{5} = \frac{2}{1577721810442023610823457130565572459346412870218046009540557861328125}$

$\frac{2}{1577721810442023610823457130565572459346412870218046009540557861328125} \times \frac{3}{5} = \frac{2}{7888609052210118054117285652827862296732064351090230047702789306640625}$

$\frac{2}{7888609052210118054117285652827862296732064351090230047702789306640625} \times \frac{3}{5} = \frac{2}{39443045261050590270586428264139311483660321755451150238513946533203125}$

$\frac{2}{39443045261050590270586428264139311483660321755451150238513946533203125} \times \frac{3}{5} = \frac{2}{197215226305252951352932141320696557418301608777255751192569732666015625}$

$\frac{2}{197215226305252951352932141320696557418301608777255751192569732666015625} \times \frac{3}{5} = \frac{2}{986076131526264756764660706603482787091508043886278755962848663330078125}$

$\frac{2}{986076131526264756764660706603482787091508043886278755962848663330078125} \times \frac{3}{5} = \frac{2}{4930380657631323783823303533017413935457540219431393779814243316650390625}$

$\frac{2}{4930380657631323783823303533017413935457540219431393779814243316650390625} \times \frac{3}{5} = \frac{2}{24651903288156618919116517665087069677287701097156968899071216583251953125}$

$\frac{2}{24651903288156618919116517665087069677287701097156968899071216583251953125} \times \frac{3}{5} = \frac{2}{123259516440783094595582588325435348386438505485784844495356082916259765625}$

$\frac{2}{123259516440783094595582588325435348386438505485784844495356082916259765625} \times \frac{3}{5} = \frac{2}{61629758220391547297791294162717674193219252742892422247678041458129878125}$

$\frac{2}{61629758220391547297791294162717674193219252742892422247678041458129878125} \times \frac{3}{5} = \frac{2}{308148791101957736488956470813588370966096263714462111238390207290649390625}$

$\frac{2}{308148791101957736488956470813588370966096263714462111238390207290649390625} \times \frac{3}{5} = \frac{2}{1540743955509788682444782354067941854830481318572310556191951036453246953125}$

$\frac{2}{1540743955509788682444782354067941854830481318572310556191951036453246953125} \times \frac{3}{5} = \frac{2}{7703719777548943412223911770339709274152406592861552780959755182266234765625}$

$\frac{2}{7703719777548943412223911770339709274152406592861552780959755182266234765625} \times \frac{3}{5} = \frac{2}{385185988877447170611195588516985463707620329643$